

Problem 1: Put 6 counters in piles of 2, 2, 1 and 1. A move goes like this: take one counter from every pile, and put those counters together as one new pile. Make moves, and after each move write the pile sizes in the next row, biggest first, like this: 4, 1, 1. Keep going until a row is exactly the same as the row above it; when that happens, the game has stopped. How many moves did it take? If you run out of rows in any table, continue on a blank sheet.

Move	Piles, biggest first
Start	2, 2, 1, 1
1	
2	
3	
4	
5	
6	
7	
8	

Moves until it stopped: _____

Problem 2: Do the same from three more starts: two piles of 3; one pile of 6; six piles of 1. Keep making moves until a row is the same as an earlier row. If it is the same as the row just above it, the game has stopped. Write the number of moves for each start.

Move	Piles, biggest first
Start	3, 3
1	
2	
3	
4	
5	
6	
7	
8	

Moves from 3, 3: _____

Move	Piles, biggest first
Start	6
1	
2	
3	
4	
5	
6	
7	
8	

Moves from 6: _____

Move	Piles, biggest first
Start	1, 1, 1, 1, 1, 1
1	
2	
3	
4	
5	
6	
7	
8	

Moves from 1, 1, 1, 1, 1, 1: _____

Problem 3: Find every way to arrange 6 counters in piles. Write each one once in the left blanks, biggest pile first: start with the biggest possible first pile and work down. How do you know you have found them all? Write your reason on the lines below. (There are 11.) Then, for each arrangement you have not run yet, run it with your counters until an arrangement repeats, and write in the right blank how many moves it took. Does every start with 6 counters end at the same arrangement?

Arrangement	Moves	Arrangement	Moves
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Problem 4: Run 4 counters from one pile of 4, and 5 counters from one pile of 5. Keep making moves until a row is the same as an earlier row. If the row it matches is not the row just above it, the game does not stop: it goes round. Then list all seven arrangements of 5 counters in the space on the right. Use your list to explain why every game with 5 counters must repeat an arrangement by move 7 at the latest.

Move	Piles, biggest first
Start	4
1	
2	
3	
4	
5	
6	
7	
8	

Move	Piles, biggest first
Start	5
1	
2	
3	
4	
5	
6	
7	
8	

Problem 5: Now 8 counters. Put all 8 in one pile. Make moves and write each row, until a row is the same as an earlier row. Did the game stop, or go round? If it went round, draw a bracket beside the rows that form the loop. How many arrangements are in the loop?

Move	Piles, biggest first
Start	8
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Stopped, or went round: _____

Arrangements in the loop: _____

Problem 6: Start 8 counters as four piles of 2, and then as piles of 5, 1, 1, 1. For each start, make moves and write rows until a row is the same as an earlier row. Which of your three 8-counter games go round on the same arrangements?

Move	Piles, biggest first
Start	2, 2, 2, 2
1	
2	
3	
4	
5	
6	
7	
8	

Move	Piles, biggest first
Start	5, 1, 1, 1
1	
2	
3	
4	
5	
6	
7	
8	

Same loop as the game from one pile of 8: _____

Problem 7: Run 10 counters from three starts: one pile of 10; two piles of 5; five piles of 2. Keep making moves until a row is the same as an earlier row. For each start, write what happened and after how many moves.

Move	Piles, biggest first
Start	10
1	
2	
3	
4	
5	
6	
7	
8	
9	

Move	Piles, biggest first
Start	5, 5
1	
2	
3	
4	
5	
6	
7	
8	
9	

Move	Piles, biggest first
Start	2, 2, 2, 2, 2
1	
2	
3	
4	
5	
6	
7	
8	
9	

From 10: _____ From 5, 5: _____ From 2, 2, 2, 2, 2: _____

Problem 8: Find an arrangement of 10 counters that does not change at all when you make a move. Is there a second one? Then try to find an arrangement of 8 counters that does not change under a move. Explain why there is none: if a move changes nothing, which pile must be the new pile, and what does that force the other piles to be?

Problem 9: Run 3 counters from three piles of 1, 7 counters from one pile of 7, and 9 counters from piles of 5 and 4, each until a row is the same as an earlier row. Then fill in the table for every number of counters from 3 to 10, using all your runs so far: does the game stop, or go round? If it goes round, how many arrangements are in each loop you found? (For 8, list both loops.)

Move	Piles, biggest first
Start	1, 1, 1
1	
2	
3	
4	
5	
6	
7	
8	
9	

Move	Piles, biggest first
Start	7
1	
2	
3	
4	
5	
6	
7	
8	
9	

Move	Piles, biggest first
Start	5, 4
1	
2	
3	
4	
5	
6	
7	
8	
9	

Counters	Stops, or goes round?	Arrangements in each loop found
3		
4		
5		
6		
7		
8		
9		
10		

Problem 10: The game stopped for 3, 6 and 10 counters, at the stairs 2, 1 and 3, 2, 1 and 4, 3, 2, 1. Build the next stairs, one step taller, with counters and count them. Then write the next three numbers of counters for which you expect the game to stop, and the stairs it should stop at.

Next stopping numbers: _____, _____, _____

Stairs they stop at: _____

Problem 11: Undo a move. Find every arrangement that turns into 3, 2, 1 after one move. Find every arrangement that turns into 5, 3, 2 after one move. Then try to find one that turns into 2, 2, 2, 2. If you cannot, explain why not.

Turn into 3, 2, 1: _____

Turn into 5, 3, 2: _____

Turns into 2, 2, 2, 2: _____

Problem 12: Sort these arrangements of 8 counters: which can be reached by a move from some arrangement, and which cannot? For each one that can, write an arrangement that reaches it. Then write what all the unreachable ones have in common, in a way that lets you decide for 6, 1, 1 and for 2, 2, 2, 1, 1, 1, 1 without running anything. Check your rule on every row of the list of 22 arrangements in Problem 13.

Arrangement	Reached from	Cannot be reached
4, 4		
3, 3, 1, 1		
2, 2, 2, 2		
2, 2, 2, 1, 1		
5, 1, 1, 1		
3, 2, 1, 1, 1		
3, 1, 1, 1, 1, 1		
4, 2, 1, 1		

Problem 13: Here are all 22 arrangements of 8 counters. Share them out at your table. Run each one until an arrangement repeats, and write L or S next to it: L if it ends in the long loop of four arrangements, S if it ends in the short loop of two. If your run reaches an arrangement someone has already run, stop: you know where it ends. Count how many arrangements end in each loop.

8	_____	4, 3, 1	_____	3, 2, 1, 1, 1	_____
7, 1	_____	4, 2, 2	_____	3, 1, 1, 1, 1, 1	_____
6, 2	_____	4, 2, 1, 1	_____	2, 2, 2, 2	_____
6, 1, 1	_____	4, 1, 1, 1, 1	_____	2, 2, 2, 1, 1	_____
5, 3	_____	3, 3, 2	_____	2, 2, 1, 1, 1, 1	_____
5, 2, 1	_____	3, 3, 1, 1	_____	2, 1, 1, 1, 1, 1, 1	_____
5, 1, 1, 1	_____	3, 2, 2, 1	_____	1, 1, 1, 1, 1, 1, 1, 1	_____
4, 4	_____				

Long loop: _____ arrangements

Short loop: _____ arrangements

Problem 14: Find a start with 10 counters that takes more than 10 moves to reach the stairs. Write each start you try in a table and record its run. You may need every row, or far fewer.

Move	Piles, biggest first
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	

Move	Piles, biggest first
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	

Problem 15: The loop for 7 counters is 4, 2, 1, then 3, 3, 1, then 3, 2, 2, then 3, 2, 1, 1, then back to 4, 2, 1. Each of these four arrangements is the stairs 3, 2, 1 with one extra counter added to one pile or added as a new pile of 1. Write all five arrangements that are the stairs 4, 3, 2, 1 with one extra counter (11 counters). Then start 11 counters at 5, 3, 2, 1 and make 6 moves. Does the game stay inside your list? Finally, write all six arrangements that are the stairs 3, 2, 1 with one extra counter on each of two different steps (a new pile of 1 counts as a step), and compare them with the two loops from Problems 5 and 6.

Stairs 4, 3, 2, 1 plus one: _____

Move	Piles, biggest first
Start	5, 3, 2, 1
1	
2	
3	
4	
5	
6	
7	
8	

Stays inside the list? _____

Stairs 3, 2, 1 plus two:

Compared with the loops for 8 counters:

Problem 16: 12 counters. Write all ten arrangements that are the stairs 4, 3, 2, 1 with one extra counter on each of two different steps. Predict how many loops there are for 12 counters and which arrangements are in each. Then test: run 5, 4, 2, 1 for 6 moves and run 5, 3, 3, 1 for 6 moves. Are they in the same loop? Find a start outside your list that ends in the loop that contains 5, 3, 3, 1.

Move	Piles, biggest first
Start	5, 4, 2, 1
1	
2	
3	
4	
5	
6	
7	
8	

Move	Piles, biggest first
Start	5, 3, 3, 1
1	
2	
3	
4	
5	
6	
7	
8	

Same loop? _____ A start outside the list that ends in the loop with 5, 3, 3, 1: _____

Problem 17: How many moves does it take from 5, 5, 5 to reach the stairs 5, 4, 3, 2, 1? And from 4, 4, 3, 2, 1, 1? Record each run. You may need every row, or far fewer.

Move	Piles, biggest first
Start	5, 5, 5
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	
23	
24	

Move	Piles, biggest first
Start	4, 4, 3, 2, 1, 1
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	
23	
24	